

# Astronautica development plan

Justifications ,initial deliverables , working method of some deliverable's, Long term development aims and budget breakdown

# Justifications

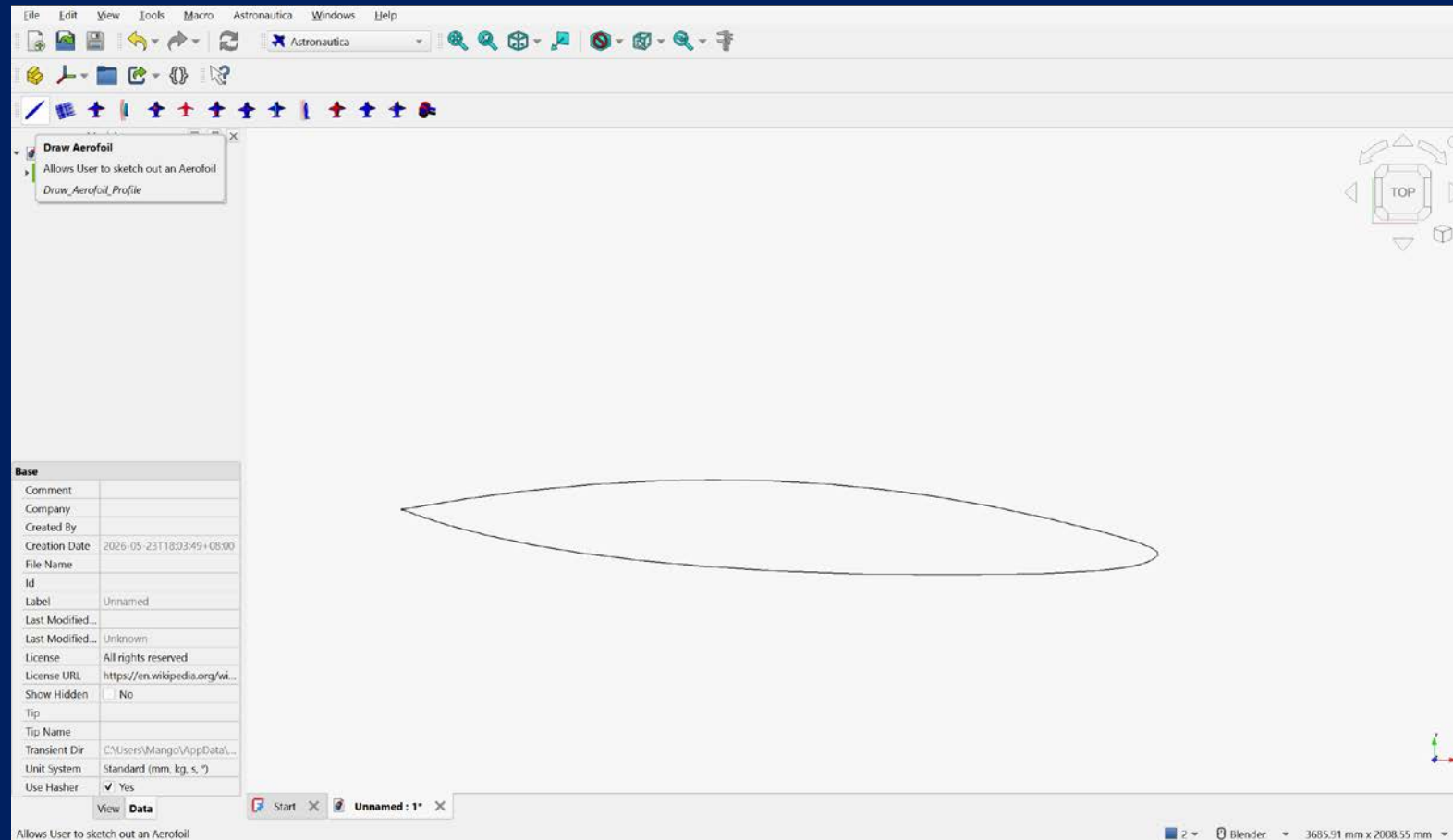
- Parametric and automated aircraft and rotorcraft design workflows have been conceptually viable for a long period of time ; however , a truly user-friendly all-in-one parametric design solution with inbuilt simulation capabilities has remained elusive since. Some examples of parametric aircraft design solutions are NASA's OpenVSP and Ntop.
- With the rapid integration of drones into the commercial sector, and the viability of concepts such as the low-altitude economy gaining significant traction, numerous startups oriented towards UAV design have formed.
- Aerospace engineers have for a long time used the industry standard CATIA for aircraft design, with Aerospace engineering students forced to use it as well.
- As a result of the rapid growth of the UAV sector and cost-reductions in equipment manufacture, hobbyist UAV's and designs are gaining traction as never before. Hobbyist users are forced to use either multiple un-intuitive solutions or commercial software once again.
- Thus , the need for a new , user-friendly all-in-one parametric aircraft design suite has been demonstrated. We believe that such a solution will see widespread adoption in a significantly short amount of time in all the aforementioned segments , bringing greater attention towards FreeCAD , spurring development.

# Initial Deliverables

- A new FreeCAD workbench designed with minimal complexity in mind utilizing existing solutions as much as possible oriented towards parametric aircraft and rotorcraft design.
- Such a workbench will have the following initial core commands ; Aerofoil selection , Wing generation with spurs placed in stress-heavy areas , Flight-control surface generation, Wing aerodynamic(CFD) simulation , Aircraft total mass calculation , Fuselage draft creation or importing , fuselage blank space definition , fuselage stringer, former and bulkhead generation , fuselage FEM simulation , fuselage CFD simulation , aircraft skin generation , aircraft rivet generation for attaching the skin to the aircraft body, plumbing for electrical routing and fuel tank systems and finally additional systems usually complex to develop , such as landing gear systems.
- In the upcoming slides , demonstration images have been used. They do not reflect the current state of the project.

# Aerofoil selection function development

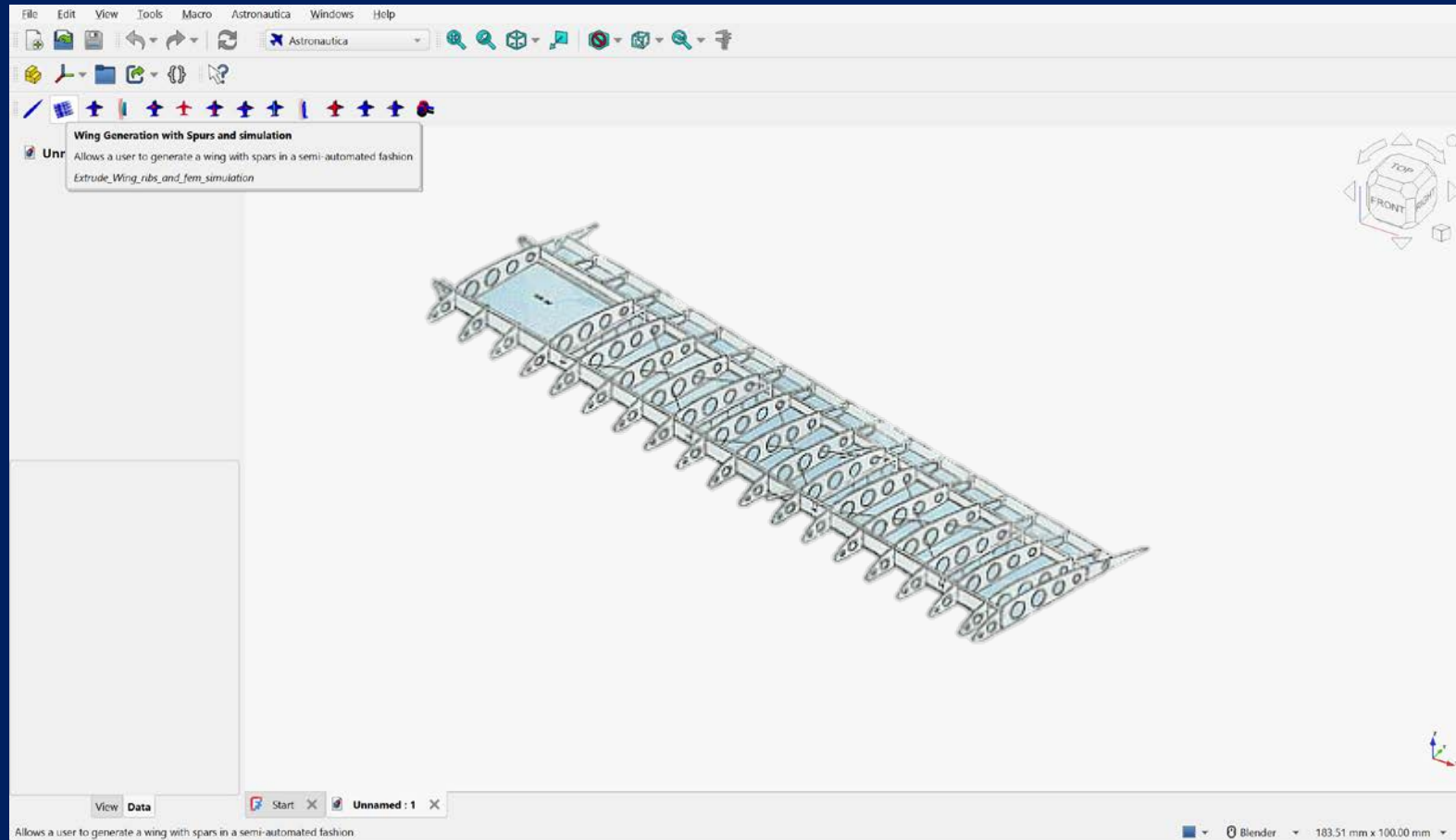
- Numerous open source FreeCAD workbenches and commands exists for the development of Aerofoil profiles . However , the aforementioned workbenches and commands are tailored only towards NACA series air foils. The TsAGI or other alternate aerofoil development schemes are not implemented. In order to ensure true widespread adoption, we must modify the existing workbenches and commands to include these .
- Primary existing examples include the “Aerofoil” workbench by Melwyn Carlo and the “Airfoil import and scale” workbench by quick61. An illustrative photo of the outcome of the Aerofoil selection function in our workbench is as follows.



# Wing generation with spurs placed in stress-heavy areas

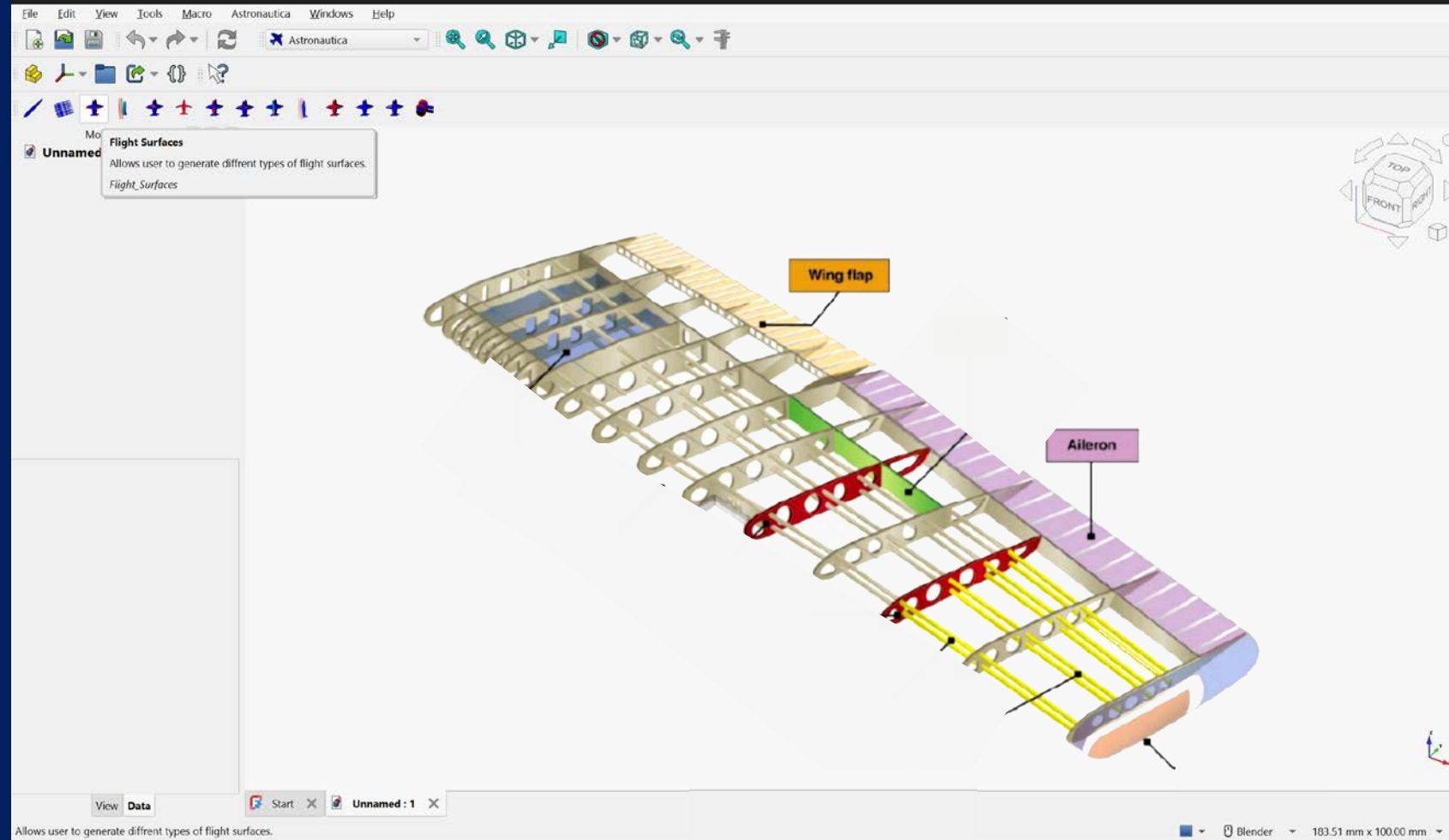
- The wing generation and spur placement function takes in user input for the wing loading and preferred length of the wing . The aerofoil profile generated in the prior step in then extruded such that it meets the length requirement defined by the user. A simulation loop begins wherein a wing loading stress is placed on various sections of the wing
- At the end of each simulation the user is prompted to place additional wing spurs on overstressed areas , after which the user resumes the simulation. This loop repeats until an ideal wing is generated , and the loop is close by the user.
- The code for this can be based on modified code from the “AirplaneDesign” workbench by FredsFactory as well as integration with FreeCad’s built in simulation functionality

Sample image in the middle : <https://www.aircraftspruce.ca/catalog/appages/despars.php?srltid=AfmBOorSdt9s93PK7VwzzAxuSgvrXHQTlWMWAwizKI4tRr2GVRufgh4Y>



# Flight-control surface generation

- The Flight surface generation feature takes in the prior wing input and adds ailerons at the user's direction . The user can define the pathway needed for the flight control rods as well as relevant control electronics
- This is the first module which must be built entirely from scratch

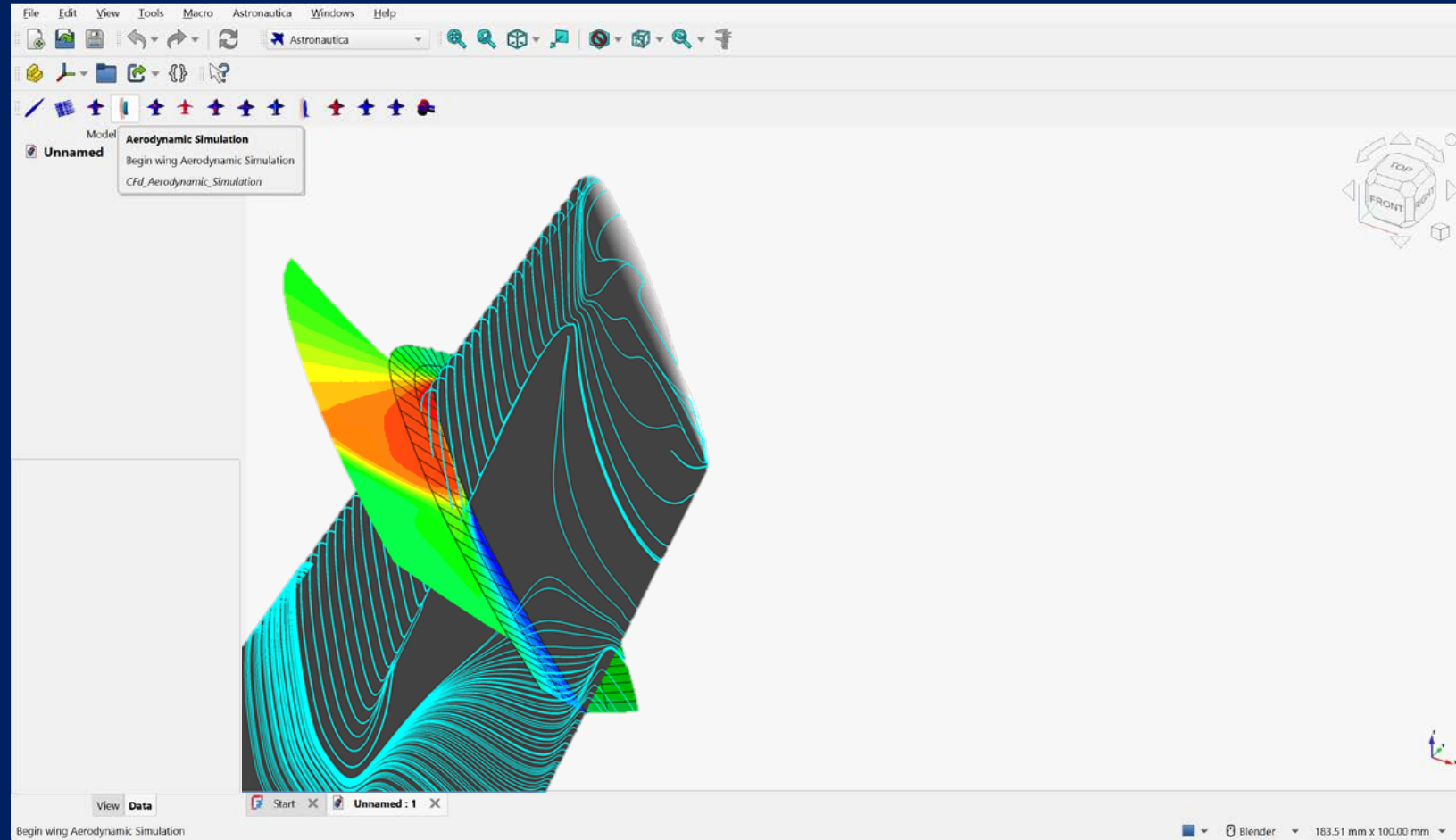


Sample image : <https://www.abbottaerospace.com/aa-sb-001/22-aircraft-specific-design-features-and-design-methods/22-16-57-wings/22-16-2-main-wing-box/>



# Wing Aerodynamics and CFD simulation

- As the name implies , this workbench will allow the user to create a wing aerodynamics simulation.
- First , a rudimentary “ideal” skin will be created by lofting the last spur to the wing tip , then the CFD simulation can begin. User will be able to control the rotation of the flight surfaces to see changes in the airflow.
- Code will mostly focus on integration with existing module's, some effort will be needed in the flight surface rotation function and UI



<https://www.onera.fr/en/news/onera-m6-wing-star-of-cfd>

# Working method of some deliverable's

- With the development procedure we will utilize for some commands demonstrated , we hope our workflow and competence in it's regard has been sufficiently demonstrated.
- We will not re-invent the wheel if not necessary , we will build upon former open-source projects and build from scratch only when it is necessary.
- We will rely on the division of labour among our research group in order to ensure that no individual member faces a large workload , whilst also ensuring that in the case of emergencies , adjunct programmers can pick up the slack.
- We will offer up to 14 bug bounties for unseen bugs we are not able to troubleshoot when user testing takes place , one for each workbench, ensuring rapid development no matter what.
- We will take in advice from professional aerospace engineers and integrate it into the module as soon as possible.



# Long term development aims

- Our module upon its prior outlined initial completion will leave a lot to be desired , as it , at that point will only support the development of fixed wing aircraft .
- Upon the completion of the aforementioned commands (hence forth referred to as fixed-wing aircraft design module) , we will immediately begin the development of a Rotorcraft module as well, similar to the workflow of the fixed wing aircraft design module.
- We will spend one year on the development of the rotor craft module and the beta testing of the fixed-wing aircraft design module.
- After we have completed the development of the rotorcraft module , we shall begin the development of a UAV module , in a similar fashion. It will introduce multi-rotor configurations.
- Perhaps our most ambitious long term goal , upon the completion of all prior goals , is the development of a Satellite module, allowing users to develop CubeSats specific to different mission requirements.
- We do not intend to halt development for a period of 5 years.

# Budget breakdown

| Item                                                                         | Cost          | Justification                                                                                                                                                     | Comment                                                                                                                                                                    |
|------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Used desktop computer (Intel Xeon , 512 GB SSD , 16 GB DDR3 RAM , GTX 750Ti) | 350-400       | A simple robust workstation that can reliably operate for the next 5 years is needed .                                                                            | Any amount of money left over will be returned by PayPal or used in other tasks with consent. Receipt will be provided . FPA can own the workstation officially if needed. |
| Website development                                                          | 300-350       | A dedicated website and forum , similar to the FreeCAD website is a must. We need to foster an independent community ; an independent developer will be hired.    | Amount left over will be sent back to the FPA or used in website upgrades.                                                                                                 |
| Website Hosting                                                              | 600-700       | A 512 GB VPS is a must , rented on a Bi-Yearly basis. We will cover the costs independently after that point.                                                     | If the FPA can provide a hosting , that would be much appreciated, rather it is preferred.                                                                                 |
| Bug Bounty(x14)                                                              | 40 per bounty | In order to ensure seamless development and community engagement , for each command , if a bug is found we are unable to fix , a 40 Dollar bounty will be offered | Any remaining amount left over after full module completion will be used in either website development of website hosting                                                  |
| Marketing                                                                    | ?             | We hope to provide seminars in our universities lecture hall on every command , any amount of money remaining can be re-invested here with the consent of the FPA | Will utilize remaining amounts from all other fundings , will be spent with FPA approval.                                                                                  |